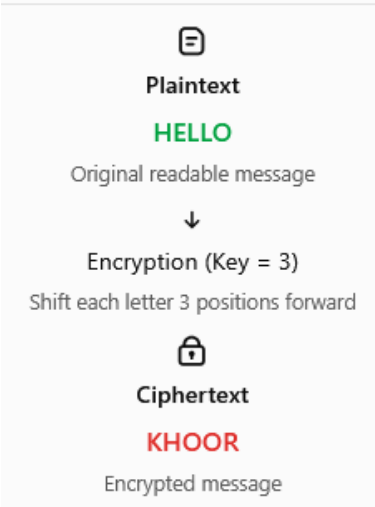
Program to implementation of Caesar Cypher Algorithm to encrypt and decrypt the string

Description: The Caesar Cipher is a simple encryption technique in which each letter in a string is shifted by a fixed number of positions in the alphabet.

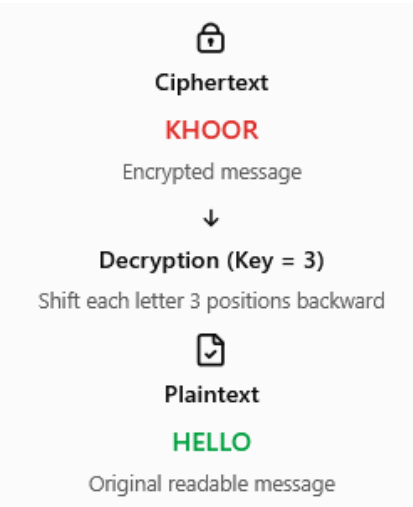
Encryption is the process of converting readable information (called plaintext) into an unreadable or coded form (called ciphertext) to protect it from unauthorized access.It uses an algorithm and a key to transform the original message into an encrypted message.



Each letter is shifted three positions forward:

- H → K
- E → H
- L → O
- L → O
- O → R

Decryption is the process of converting encrypted, unreadable information (called ciphertext) back into its original readable form (called plaintext).Decryption uses an algorithm and a key to recover the original message.



Each letter is shifted three positions backward:

- K → H
- H → E
- O → L
- O → L
- R → O

Algorithm:

Encryption of a letter by a shift n can be described mathematically as. $E_n(x) = (x + n) mod\ 26$

Decryptio of a letter described mathematically as $D_n(x) = (x - n) mod\ 26$

- X = Position of the plaintext letter
- $E_n(x)$= Position of the ciphertext letter
- n = Secret key

- 26 = Number of letters in the English alphabet

Encryption

1. Read the input string and the key.
2. Traverse each character of the string.
3. If the character is an uppercase or lowercase letter, shift it forward by the key positions.
4. Preserve spaces and special characters.
5. Display the encrypted string.

Decryption

1. Traverse each character of the encrypted string.
2. Shift each alphabetic character backward by the key positions.
3. Preserve spaces and special characters.
4. Display the decrypted string.

Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main()
{
   char text[100], encrypted[100], decrypted[100];
   int key, i;

   printf("Enter a string: ");
   fgets(text, sizeof(text), stdin);

   printf("Enter the key: ");
   scanf("%d", &key);

   key = ((key % 26) + 26) % 26;

   // Encryption
   for (i = 0; text[i] != '\0'; i++)
   {
      char ch = text[i];

      if (isupper(ch))
         encrypted[i] = (ch - 'A' + key) % 26 + 'A' ;
      else if (islower(ch))
         encrypted[i] = (ch - 'a' + key) % 26 + 'a' ;
      else
         encrypted[i] = ch;
   }
   encrypted[i] = '\0';
```

```
    printf("Encrypted string: %s", encrypted);

    // Decryption
    for (i = 0; encrypted[i] != '\0'; i++)
    {
        char ch = encrypted[i];

        if (isupper(ch))
            decrypted[i] = (ch - 'A' - key + 26) % 26 + 'A';
        else if (islower(ch))
            decrypted[i] = (ch - 'a' - key + 26) % 26 + 'a';
        else
            decrypted[i] = ch;
    }
    decrypted[i] = '\0';

    printf("Decrypted string: %s", decrypted);

    return 0;
}
```

OUTPUT:

```
Enter a string: HELLO WORLD
Enter the key: 3
Encrypted string: KHOOR ZRUOG
Decrypted string: HELLO WORLD
```